

JEREMIAH NYAGAH TECHNICAL INSTITUTE

ELECTRICAL & ELECTRONIC ENGINEERING DEPARTMENT

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18-10-2021

Final Circuits

Definition

A final sub-circuit is an outgoing circuit connected to a distribution board and intended to supply electrical energy to current-using apparatus, either directly or through socket outlets or fused spur-boxes.

Examples of final sub-circuits includes; lighting, socket-outlets, cooker and water heater. A final sub-circuit originates from the consumer control unit (CCU) or distribution board (DB). Each final sub-circuit is protected by an appropriate fuse or circuit breaker mounted in the consumer control unit.

IEE Regulations

- i) Where an installation comprises more than one final sub-circuit, each shall be connected to a separate way in a distribution board.
- ii) The wiring of each final sub-circuit shall be electrically separate from that of every other final sub-circuit to facilitate disconnection of each final sub-circuit for testing.
- iii) Every final sub circuit shall have means of protection against excess current
- iv) The neutral conductor shall never be connected in the same order as that in which the live conductors are connected to the fuses or circuit breakers.
- v) The neutral conductor shall never be connected to fuses, switches or circuit breakers.

Requirements for the consumer intake point

The IEE regulations requires that, every consumer's installation shall be adequately controlled by a switchgear readily accessible to the consumer which shall incorporate-

- i) Means of isolating all the conductors of the installation of the premises from the supply.
- ii) Means of excess current protection.
- iii) Means of earth leakage protection.

Sequence of control

The sequence of the equipment forming the switchgear required shall be such that the means of isolation follows the consumer's terminals without the intervention of any other apparatus as shown in figure 24

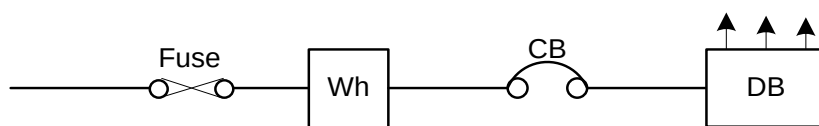


Figure 1 Sequence of control

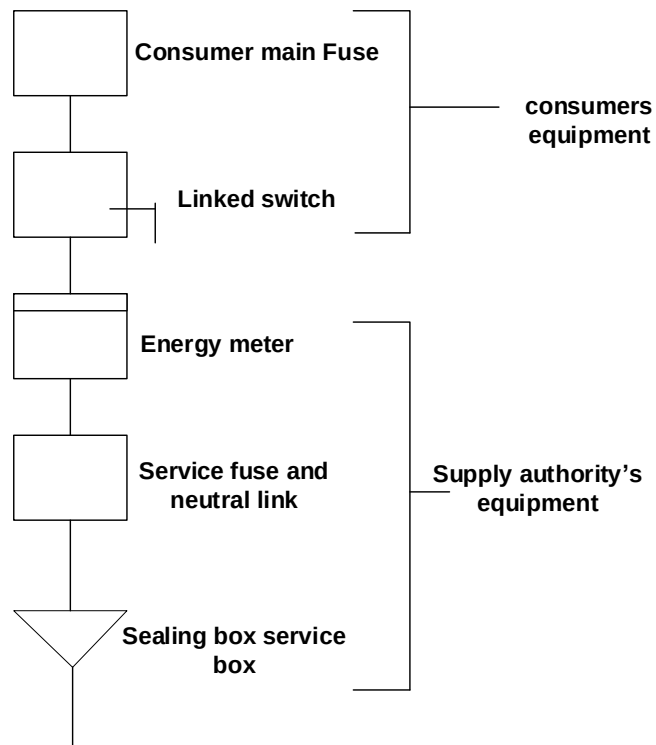


Figure 2 Single line diagram for the equipment at the consumer intake point

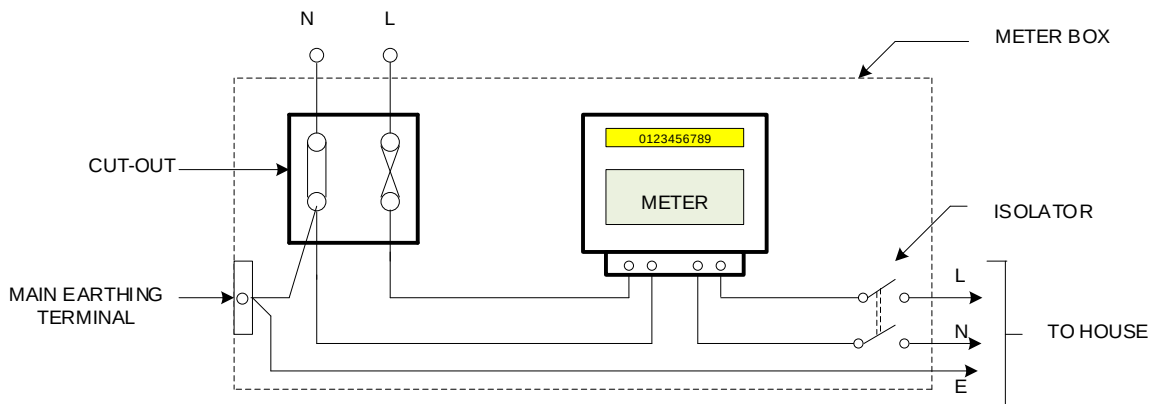


Figure 3 Circuit diagram for the equipment at the consumer intake point

Meter boxes can be mounted on the wooden /masonry surfaces or chased walls. When mounting a meter box, make sure that it is at the correct height which is reachable by a person standing on the ground. This is to make it easily accessible for energy meter reading, any repairs and replacements.

I.E.E regulations

- i) Every consumer's installation shall have a means of isolation
- ii) Every consumer's installation shall have a means of excess current protection.
- iii) Every consumer's installation shall have a means of earth leakage protection.
- iv) Where a consumer's installation comprises installation in two or more detached buildings, separate means of isolation shall be provided.

- v) Every means of excess current protection shall be suitable for the maximum short-circuit current attainable.

Need For Switching

- i. Used to put a circuit in use or out of use.
- ii. Emergency switching.
- iii. Isolating a circuit from the supply for maintenance.

Types of switches

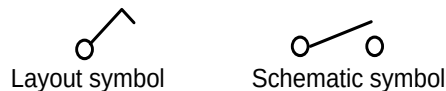
1. Single pole switch
2. Double pole switch
3. Triple pole switch.

I.E.E regulation on switches

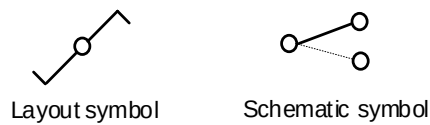
- i) All single pole switches should always be connected in the live (phase) conductor only.
- ii) Fuses shall also be inserted in the live only.

Types of single pole switches

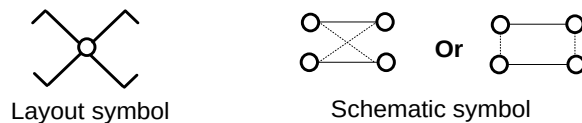
1. One-way switch - This has one path only for current.



2. Two-way switch - This has two alternative paths for current (change over switch).



3. Intermediate switch – This has two double alternative current paths.



Switching of lighting points

One-way switching.

In the one-way switching, the lamp or set of lamps is switched from one particular point. An example is the switching at the entry of a room with one entry only.

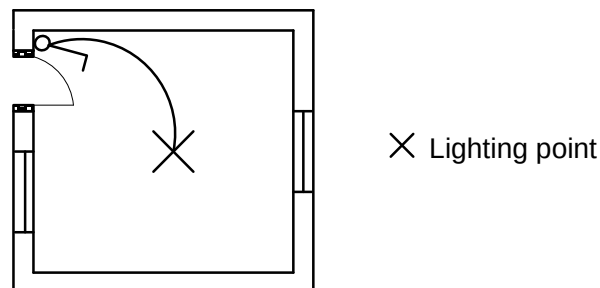


Figure 4 Layout of one room with one-way switching arrangement.

The ON and OFF control of the lighting point is only at one position.

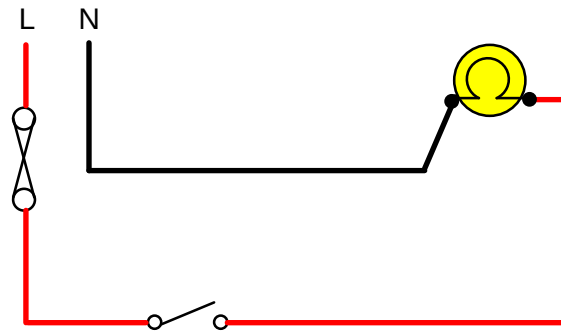


Figure 5 Schematic diagram showing one-way switch controlling one lamp.

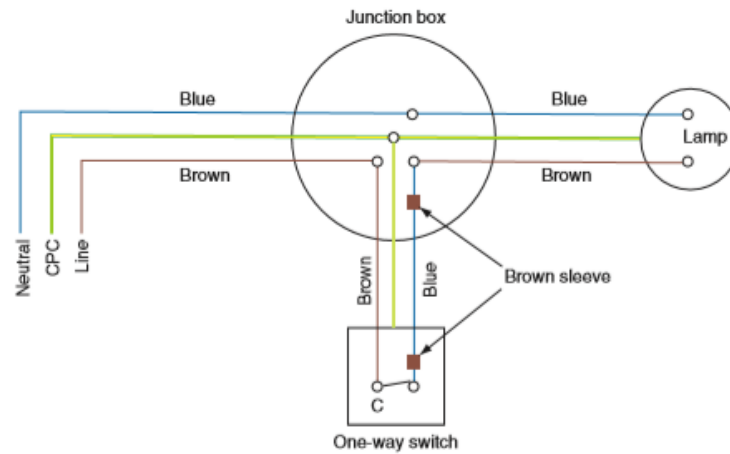


Figure 6 One way switching using a joint box

Two – Way Switching

Where it becomes necessary to have the lamp or set of lamps switched from two separate positions. An example is a long corridor or in a room with two entries.

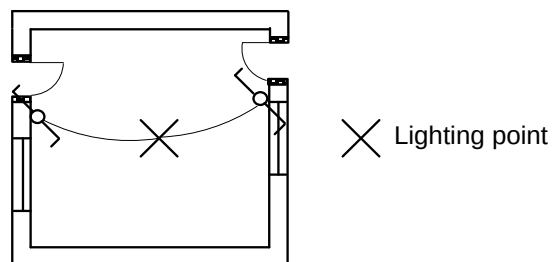


Figure 7 Layout of two-way switching arrangement.

The ON and OFF control of the lighting point is from two positions.

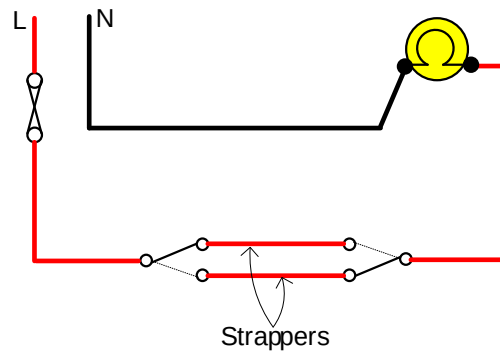


Figure 8 Schematic diagram showing two-way switch controlling lighting point.

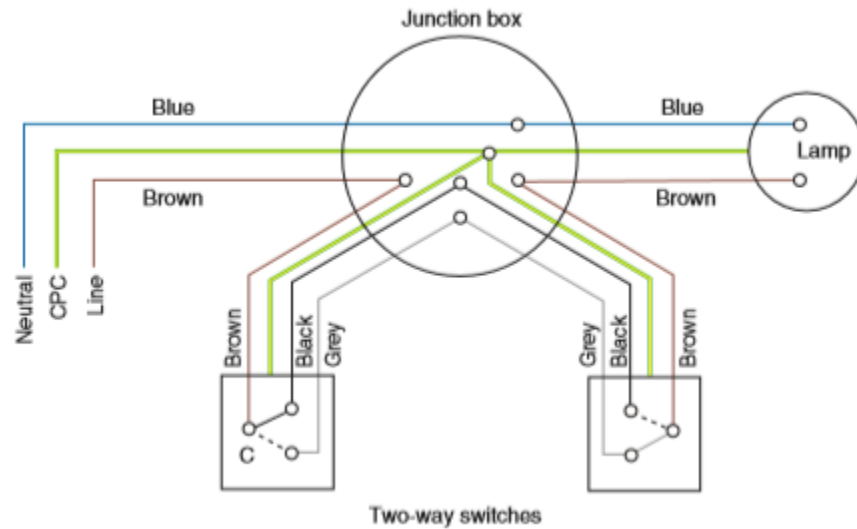


Figure 9 Two way switching using a joint box

Intermediate switching

This is used in conjunction with the *Two-way* switching where the switching of the lamps is from more than two points. All the other points of switching between the two two-way switches will be fitted with intermediate switches. An example of lamp controlled from THREE positions is shown in figure 33.

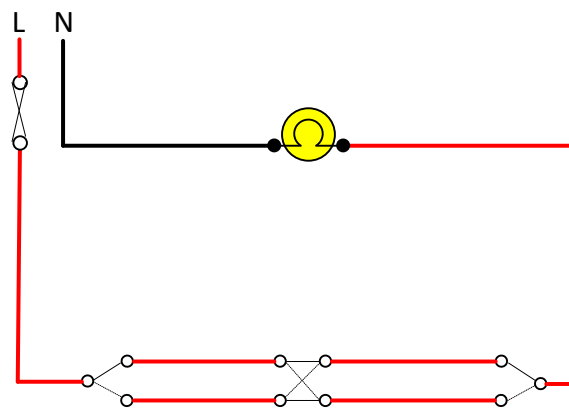


Figure 10. Intermediate switching

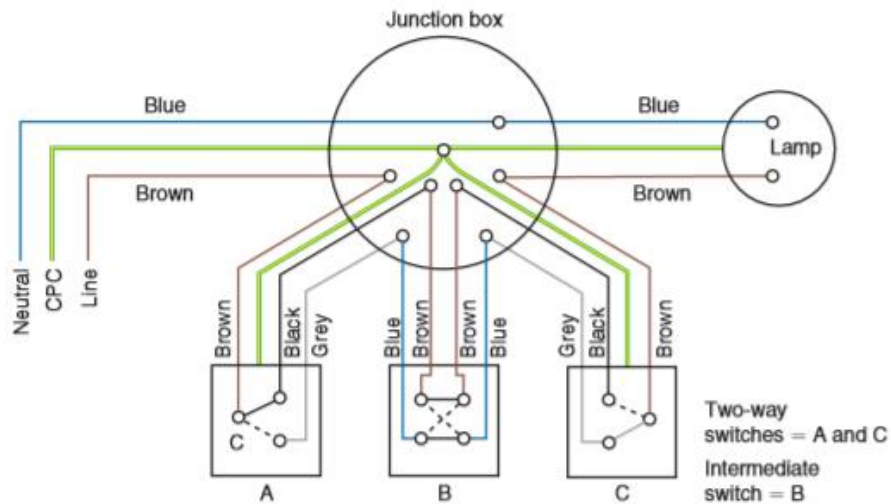


Figure 11 Intermediate switching using a joint box

Note

A lamp or set of lamps may be controlled from any number of positions provided we have 2 two-way switches and the rest of the switches been intermediate

Types of circuit connections

1. Series connection
2. Parallel connection

Series Connection

The current using equipment are connected in series or one after the other. One end of the equipment is connected to the other end (*End to End*) in series connection, the current flowing is the same and the voltage across each equipment will depend on the rating and its resistance

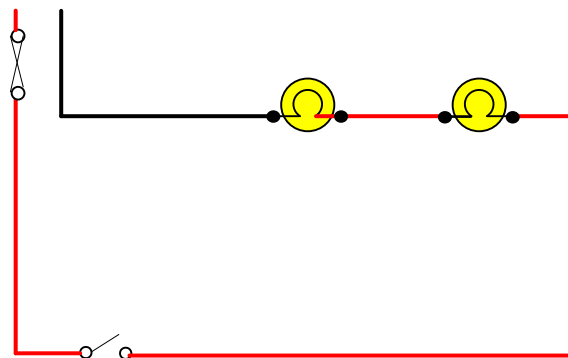


Figure 12 Series connection

The supply voltage is dropped individually across each lamp and depending on the ratings the voltage across each equipment will vary but the current flowing through all the equipment is the same. That is, if two lamps of equal resistance are connected in series to a 240 V supply, the voltage which will appear across each lamp will be 120 V and the current will be the same in all lamps

N.B

The lamps in above example may not produce light or be dim because the light produced is equal to

$$Power = V^2/R$$

And, V is below rated voltage

For a series connection to work, all lamps have to be good working condition in order to provide continuity of the circuit.

Disadvantages of series

1. If one lamp blows, all lamps go off.
2. Lamps will be dim (will not produce enough light) or may fail to light
3. Not reliable- Therefore for practical purposes series circuits are not used

Parallel connection

This is the most common arrangement for lighting circuits. All lamps are connected across the supply and each lamp receives the supply voltage across it. Each lamp can be controlled separately. If one lamp blows out, all the other will continue working.

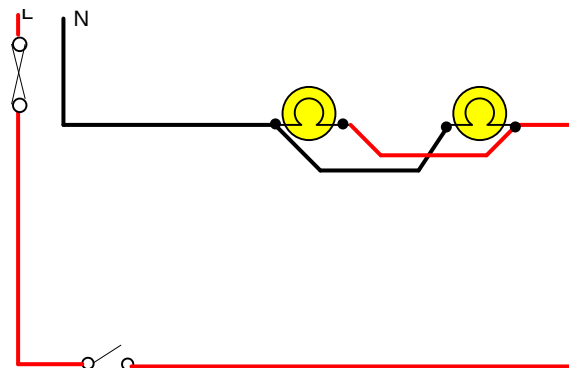


Figure 13 Parallel connection

Note:

If separately controlled, all switches are to be connected in the live conductor and that no wire goes directly to the lamp bypassing the switch.

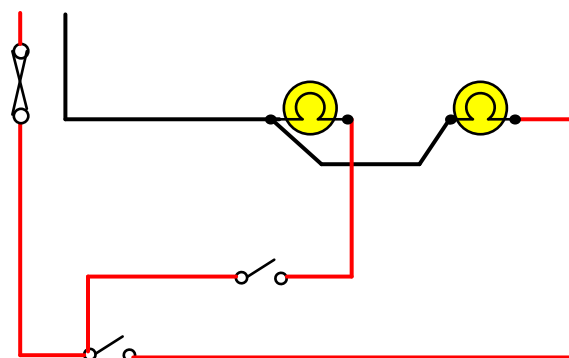


Figure 14. Separately controlled lamps

Advantages of parallel

- i) All lamps receives rated voltage
- ii) All lamps are bright
- iii) If one lamp fails , other will not be affected
- iv) Lamps can be controlled individually (separately)

Looping-In System

Looping-in of wires helps in saving materials and provides a clean piece of work. Looping – in can be done at:

- i) Switches
- ii) Lamp Holders
- iii) Junction Boxes
- iv) Ceiling Roses

If more than one lamp are to be switched from the same switch, it becomes cheaper to loop-in at the lamp holder for the second lamp. Figure 38 shows the theoretical diagram of a final sub-circuit of seven lamps, two controlled separately by 1-way switches, three controlled as a group by a 1-way switch, and two controlled by a 1-way switch. If the circuit were to be wired exactly as in the diagram, a large number of joints would be necessary. Figure 39 shows the same circuit as wired by the looping-in system. No joints are required except dry twisted joints in the terminals of the two-plate ceiling roses and of the single-pole one-way switches.

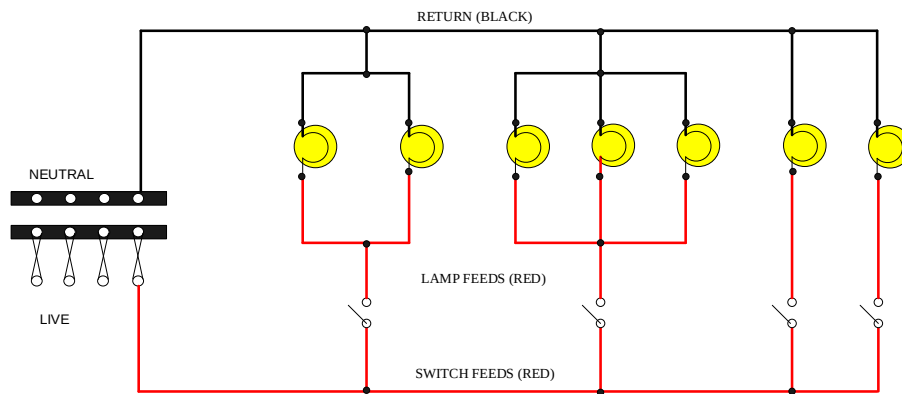


Figure 15 Control of lamps

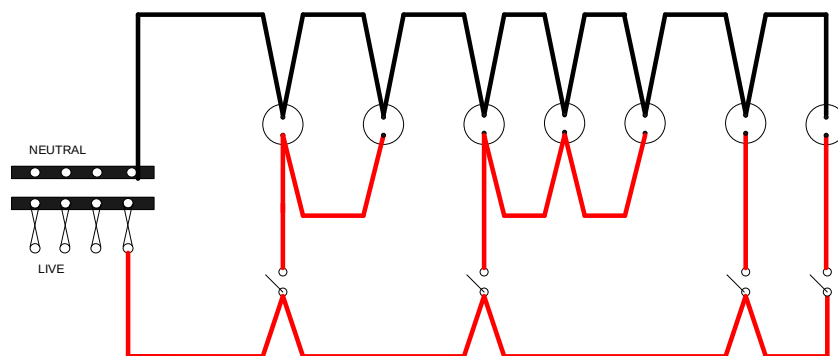


Figure 16 Looping in system

Looping in from a switch.

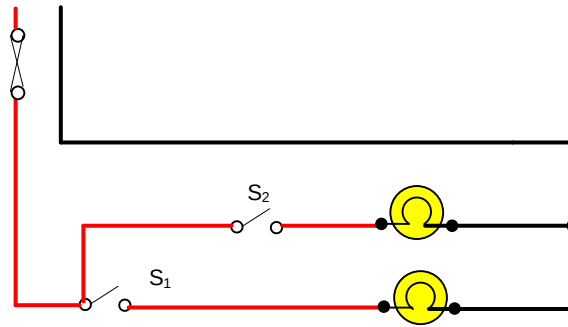


Figure 17 looping in from switch

In this system the Live to switch S_2 is normally looped from S_1 .

Looping-in from a three plate ceiling rose

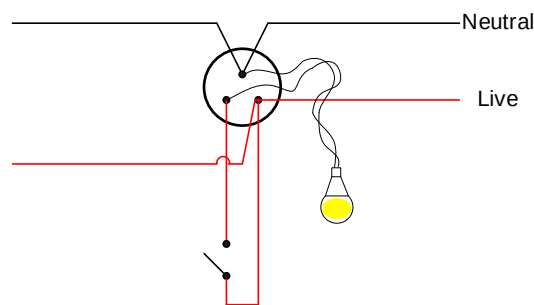


Figure 18. Looping in from ceiling rose

A ceiling rose must not be connected in such a manner that one terminal remains alive when the associated switch is OFF, unless that terminal cannot be touched when the ceiling rose is partially dismantled to allow flexible cord replacement.

Looping-in at Junction Boxes

In this method, all the circuit wires are brought to a common Box and distributed to the switches, ceiling roses, lamp holder's e.t.c.

This method of looping in is only used where there are few lamps and more so where there is service wiring using twin core cables.

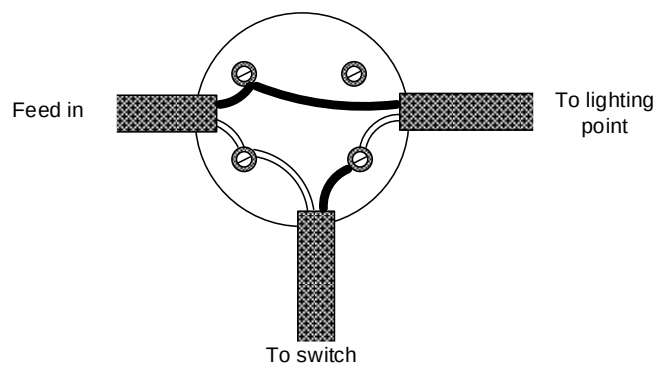


Figure 19 Looping in at joint box

Colour identification of cables

Single phase

Live - Red
Neutral- Black
Earth - Green or green /yellow

Three Phase

L₁ - Red - phase
L₂ - yellow - phase
L₃ - blue - phase
Neutral- black

Size of cables and rating of protective devices

Cables are manufactured in different sizes and compositions and each is designed to carry a maximum amount of current. The current a cable can carry are given in the I.E.E Regulation tables of current ratings. For lighting circuits the cable commonly used is the 1mm^2 or 1.5mm^2 which is rated 18A depending on the type of cable. The most common cable used is single core and Twin flat and Twin with earth. Most lighting circuits will be rated at either 5A or 10A because most switches are rated 5A or 10A.

The I.E.E Regulations gives the protective ratings of different types of lamps as shown below.

Type of Holder	Maximum Rating of fuse or Circuit Breaker.
Small Bayonet type	5A
Small Edison –type screw	5A
Bayonet type	15A
Bi- pin type	15A

I.E.E Regulations on switches

1. Every switch or Circuit Breaker the purpose of which is not obvious shall be labeled to indicate the apparatus it controls.
2. All single pole switches shall be always connected to the live conductor.
3. Every switch or other electric control shall be placed so as to be out of reach of a person in contact with bath, shower unit etc.
4. In a two wire installation connected to a supply having neither pole connected with the earth, switch or circuit breaker shall be of double pole linked type and the fuses shall be installed in both poles.
5. In a two wire installation connected to a supply having one pole earthed, switches shall be connected in the live conductor only.

Power Circuits

Final power sub-circuits

A final power sub-circuit is a circuit that allows electrical appliances to access electrical power.

All final sub-circuits must be electrically separate that is there must be no “bunching” of neutral conductors. All neutral conductors must be connected at the distribution board in the same order as the line conductors.

Socket Outlet

A socket outlet is a device provided with female contacts which is intended to be fitted with the fixed wiring and intended to receive a plug. A device with protected current carrying contacts intended to be mounted in a fixed position and permanently connected to a fixed wiring of a an installation to enable the connection to it of a flexible cord or flexible cable by means of a plug, or is an accessory with 3 terminals marked *L*- live, *N*- Neutral and *E* – Earth fitted with the fixed wiring ready to receive a plug.

Plug



This is a device intended for connection to a fixed cord or flexible cable which can be engaged manually with a socket outlet and which has current carrying contact pins which may be exposed when not engaged. Socket outlet will be wired such that the terminal marked *L* will receive the live wire, the terminal marked *N* will receive the Neutral and the one marked *E* will receive the earth wire. Socket outlets must be installed in a place to wait for a plug but not the vice-versa

Note;

Sometimes the earth terminal is marked with the Earth symbol

There are two ways in which socket outlets may be wired

1. Radial circuit
2. Ring circuit

Radial Circuit

A radial circuit is a circuit in which the live, neutral and earth conductor start from the distribution board to sockets connected in series and terminate at the last socket.

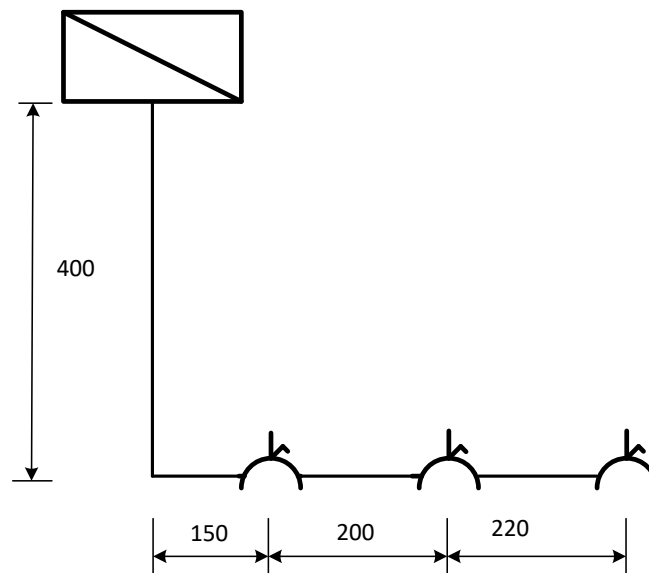


Figure 20. Layout diagram of a radial circuit

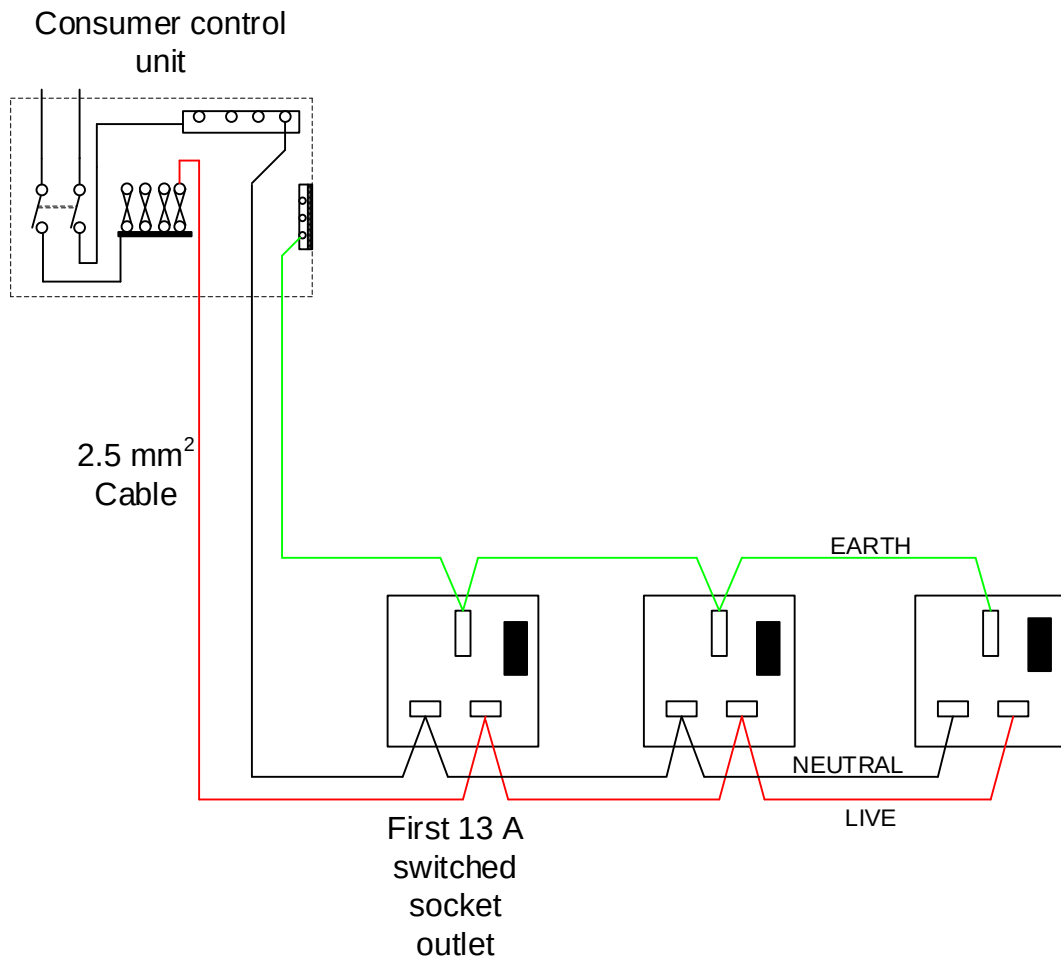


Figure 21 Wiring diagram for the radial circuit

Ring Circuits

A ring circuit is defined in the I.E.E. Regulations as “a final sub-circuit in which the current-carrying and the earth-continuity conductors are connected in the form of a loop, both ends of which are connected to a single way in a distribution board or its equivalent.

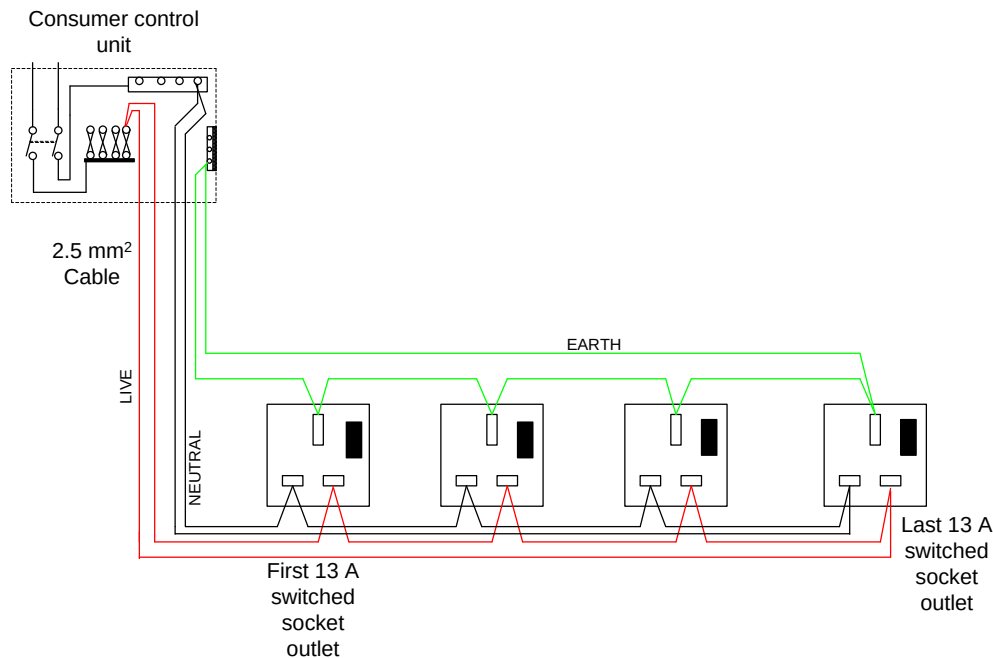


Figure 22. Wiring diagram for a ring circuit.

Spur

A spur is a socket that branches off a ring circuit. The number of spurs fed from a ring circuit must not exceed the number of sockets in the ring. The cables connecting the spur must be of the same size as that of the ring circuit. The minimum cable size for ring and radial circuit is 2.5mm^2 .

Advantages of a ring circuit over a Radial circuit

1. There is total safety in ring circuit than in Radial for the circuit is in form of a ring
2. An open circuit point in the Ring circuit will not affect any other socket in the system
This is because there are two paths of current and if one path is open the current will flow through the other
3. Smaller sizes of cables may be used in Ring than in a Radial. This is because the Ring has two parallel current paths
4. More loads may be fed or connected to a Ring circuit than a Radial circuit of the same capacity

Disadvantages of Ring over Radial

1. More cables lengths requires in a Ring than in Radial therefore becomes more expensive
2. It consumes more time to install a Ring circuit than Radial because more cables has to be installed.

I.E.E Regulations on Ring and Radial circuit

1. A Ring circuit it may serve an unlimited number of points but shall not serve an area of more than 100m^2

2. For a Ring final sub-circuit the total number of spurs shall not exceed the total number of socket outlets and stationary appliances connected directly in the ring circuit.
3. Where two or more ring final sub circuit are installed, the socket outlet to be served shall reasonably be distributed among the separate ring circuit
4. For a Radial or Ring final sub circuit the Rating of the fuse or circuit breaker at the circuit breaker shall not exceed 30A
5. Each circuit conductor of a ring final sub circuit shall be run in the form of a Ring commencing and returning to the same way in the circuit breaker.
6. Except where Ring is run throughout in metallic conduct, Ducts or Trunking, Earth continuity conductor shall run in form of a ring originating and returning to same point in the circuit breaker.
7. For a Radial circuit the current rating of the circuit conductors shall not be less than Rating of the fuse or circuit breaker protecting the final sub- circuit

Electric Bell

There are various types of electric bells including the single stroke bell, the trembles, the buzzer and a continuously ringing bell, but all depend on the attraction exerted by the electromagnet or a soft iron armature.

i) Single stroke bell

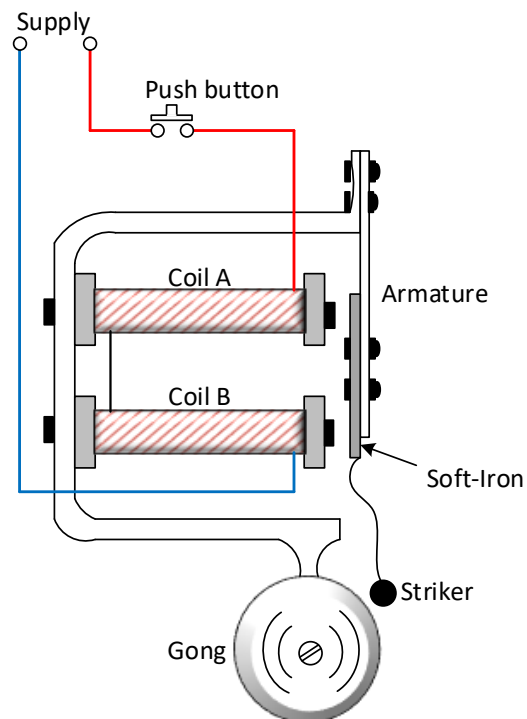


Figure 23 Single stroke bell

Construction

The bell is made up of two coils wound up with insulated copper wire on an insulated bobbin. The bobbins are slipped on to a U-shaped soft iron core and the magnetic circuit is completed by a soft-iron strip riveted to a spring-loaded armature. The striker is attached to the armature.

Operation.

1. the push-button is pressed
2. Current passes through the coil.
3. The coil is magnetized.
4. The soft-iron strip on the armature is attracted towards the coil.
5. The armature carries a striker which hits the gong.
6. The gong produces audible sound.
7. When the circuit is broken the coil becomes demagnetized and the spring pulls the armature back into its original position.
8. The striker will only operate when the push is operated hence a single stroke.

ii) **Trembler bell**

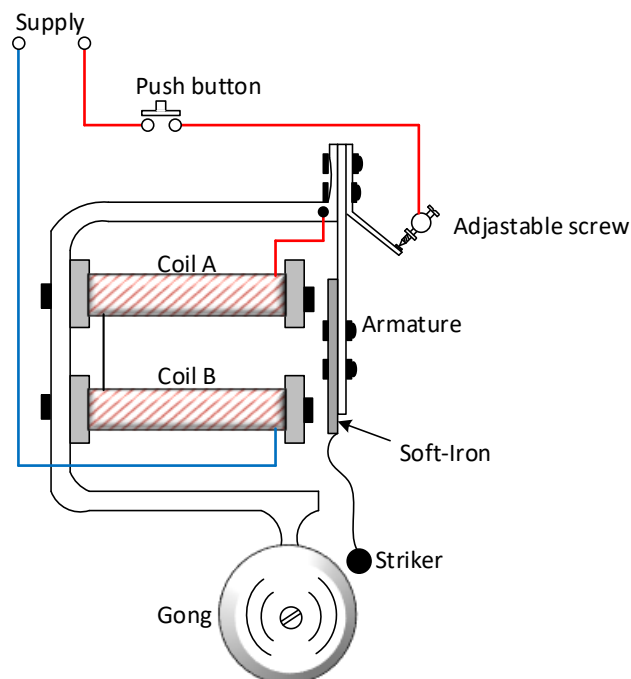


Figure 24 Trembler bell

Construction.

This type of bell is similar in construction to the single-stroke bell except that there is an adjustable pair of contacts between the armature and the push button contact.

Operation.

1. The push-button is pressed
2. Current passes through the coil.
3. The coil is magnetized.
4. The soft-iron strip on the armature is attracted towards the coil.
5. The armature carries a striker which hits the gong.
6. The gong produces audible sound.
7. When the circuit is broken the coil becomes demagnetized and the spring pulls the armature back into its original position.
8. The cycle is then repeated, giving a continual make-and-break action.

Continuous ringing bell

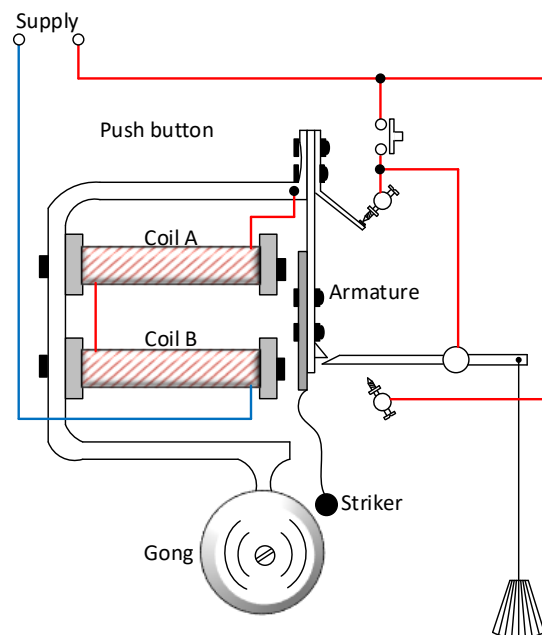


Figure 25 Continuous ringing bell

Construction

The continuous ringing bell differs in construction from the trembler bell in that a small lever is placed below the contact screw. This lever drops automatically on the first stroke of the bell, and as it drops it shorts out the bell-push, thus causing the bell to ring continuously until the lever is reset by the cord. Confusion can arise in the drawing of this circuit if the operation of the lever is not understood.

Operation.

1. When the bell –push is operated the coils become an electromagnet and the armature is drawn towards them.
2. The lever drops on the first stroke and shorts the push.
3. The bell now operates as a trembler bell.
4. To stop the bell, pull the cord, thus resetting the lever.

iii) **The Buzzer**

The buzzer is a trembler bell without a striker and gong. The principle of operation is similar to that of the trembler bell, however, the sound it produces is like the buzz of a bee.

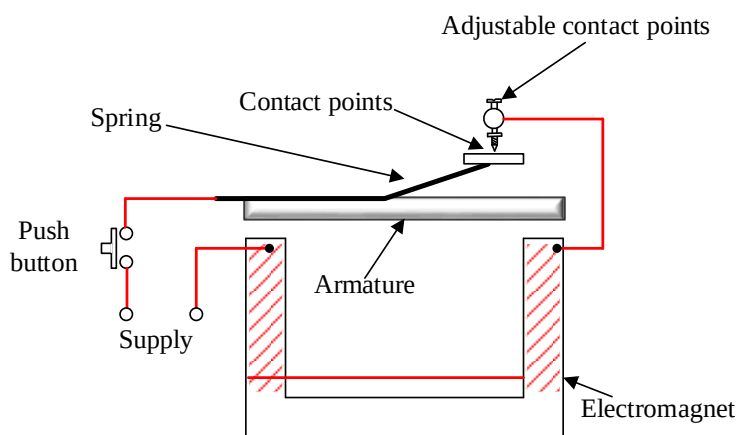


Figure 26 Buzzer

Bell Indicators

These are devices used to indicate the location where the bell is being rang by showing in the indicator board. These indicators coils are connected in series with the bell Indicator boards are situated at places where the intended person can access it. For example, it can be situated in the nurses' office to show when a patient is in need. During bell circuit installation comprising more than one bell push buttons operating the same bell, it is necessary to include an indicator board which will show from which push button the bell has been rung. There are three types of bell indicators.

- 1) Pendulum
- 2) Electrical replacement.
- 3) Mechanical replacement.

1. Pendulum type

In the pendulum type there are movements similar to that of the single stroke bell where a soft iron armature carrying a flag is pivoted at its end in front of an electromagnet. The coils of which are

in series with the push concerned .When the push button is pressed, the electromagnet attracts the armature. When press the button is released, the electromagnet is demagnetized releasing the armature .The armature starts to swing in a pendulum motion before resting.

Disadvantages

- i. When the person called happens to be away from the indicator board when the bell rings, the pendulum may stop swinging by he/she comes back.
- ii. There may be confusion if more than one push buttons are pressed almost at the same time.

2. Electrical replacement type

In this arrangement, the element consists of two electromagnets, one of which is in series with the bell circuit. The second coil is connected in the replacement circuit. The armature is pivoted at its center about which it will rock. The flag arm is attached to the armature. When the current flows through the alarm circuit, one end of the armature is attracted, causing the flag to overbalance and fall to one side. To restore the flag, the current is passed through the replacement coil, which restores the flag to its original position.

3. Mechanical replacement type

Has an armature which is attracted by the electromagnet to show the flag in its appropriate space in the screen or window of the indicator board. The flag remains in this position even though the bell push button is released until it is reset by a lever that is pushed or turned around by hand.

Relays

This is an electrical device such that current flowing through it in one circuit can switch ON and OFF current in a second circuit.

Construction

Basic parts and functions of electromechanical relays include:

1. **Frame:** Heavy-duty frame that contains and supports the parts of the relay.
2. **Coil:** Wire is wound around a metal core. The coil of wire causes an electromagnetic field.
3. **Armature:** this is the relays moving part. The armature opens and closes the contacts.
4. **Contacts:** The conducting part of the switch that makes (closes) or breaks (opens) a circuit.
5. An attached **spring** returns the armature to its original position.

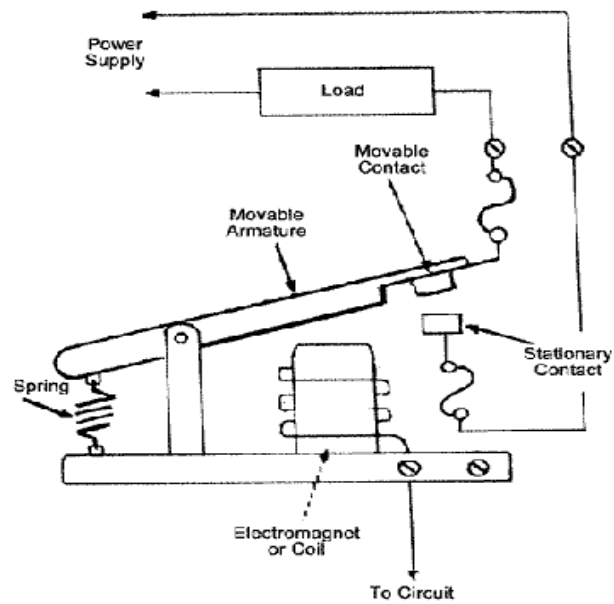


Figure 27 Electromechanical relay

Bell circuits

There are layout diagrams and wiring diagrams which use BS 3939 symbols. An example of a layout diagram is shown in figure 51.

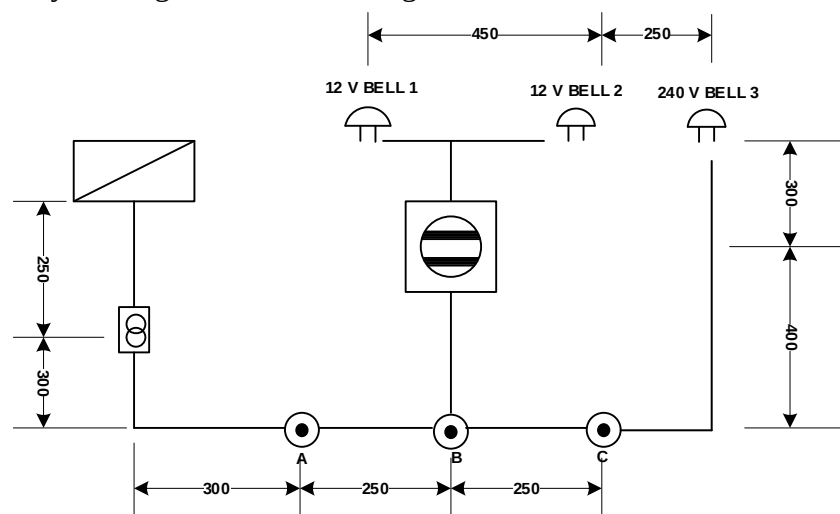


Figure 28 layout diagram for alarm circuit

Description

Figure 51 shows a bell circuit layout diagram, push buttons A and B controls bells 1 and 2 simultaneously. The bells are connected in parallel, and push button C controls bell 3. Bells A and B are connected through an indicator board.

Figure 52 shows the wiring diagram of the layout diagram in figure 51

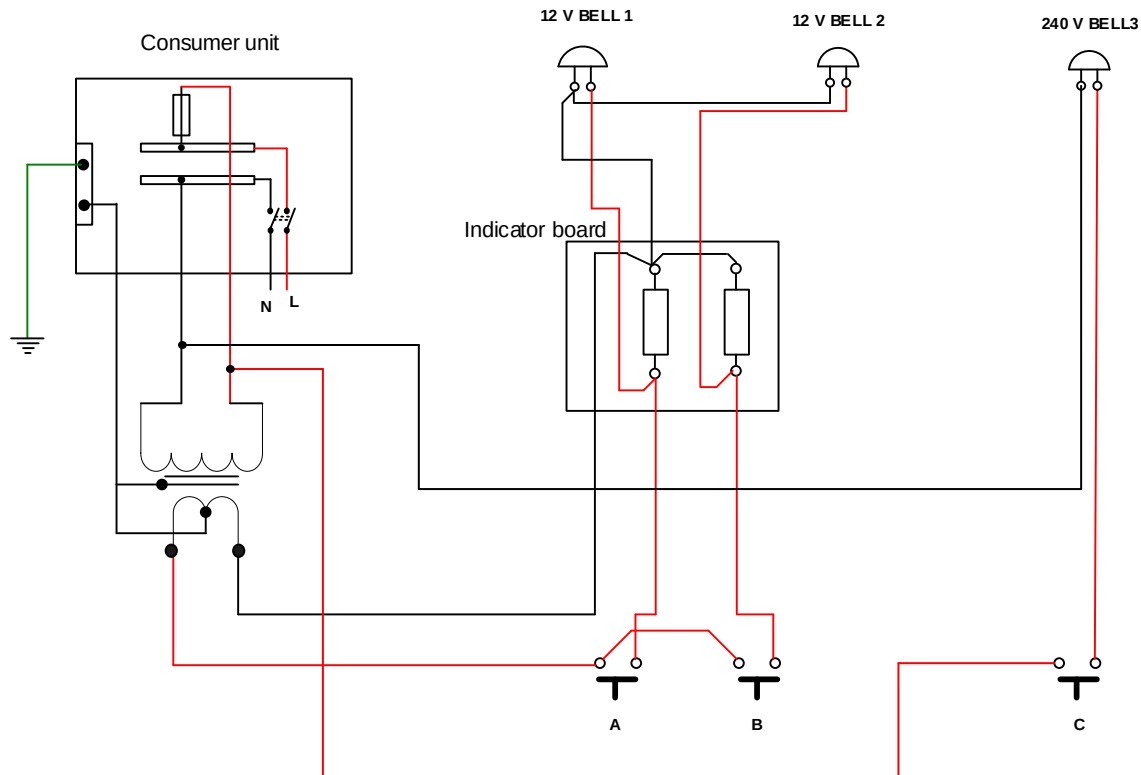


Figure 29 Bell circuit wiring diagram

ELECTRIC WATER HEATERS

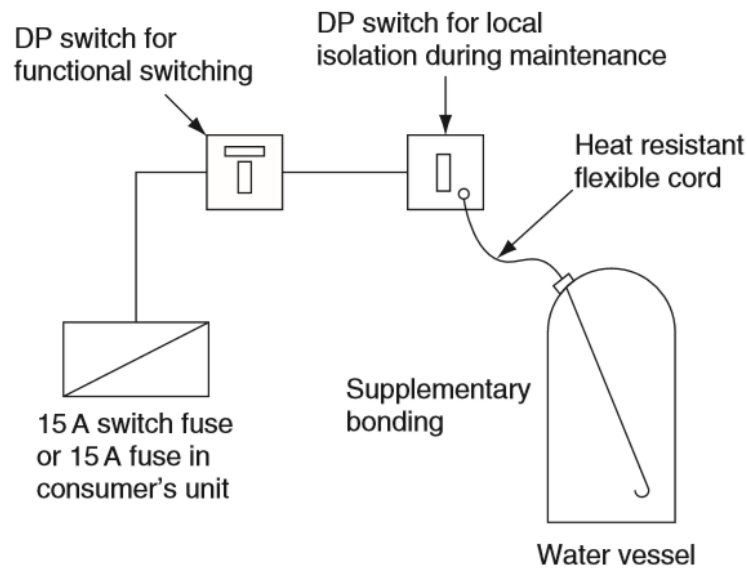


Figure 30. Water heater layout diagram.

There are two basic types of electric water heaters;

- (a) The free outlet or non- pressure-type (N.P.T) and
- (b) The pressure type.

Free Outlet Water Heater

Both types of water heater operate on the principle that water, when heated, becomes less dense (lighter) and rises. An immersion heater element is fitted at the bottom of the tank or vertically (through the center) so that all the water in the tank will be heated. The immersion heater will only heat the water above it. But in the free outlet type cold water is fed into the tank through an inlet valve. When this valve is opened, the incoming cold water pushes the less dense hot water in to the outlet pipe.

Construction

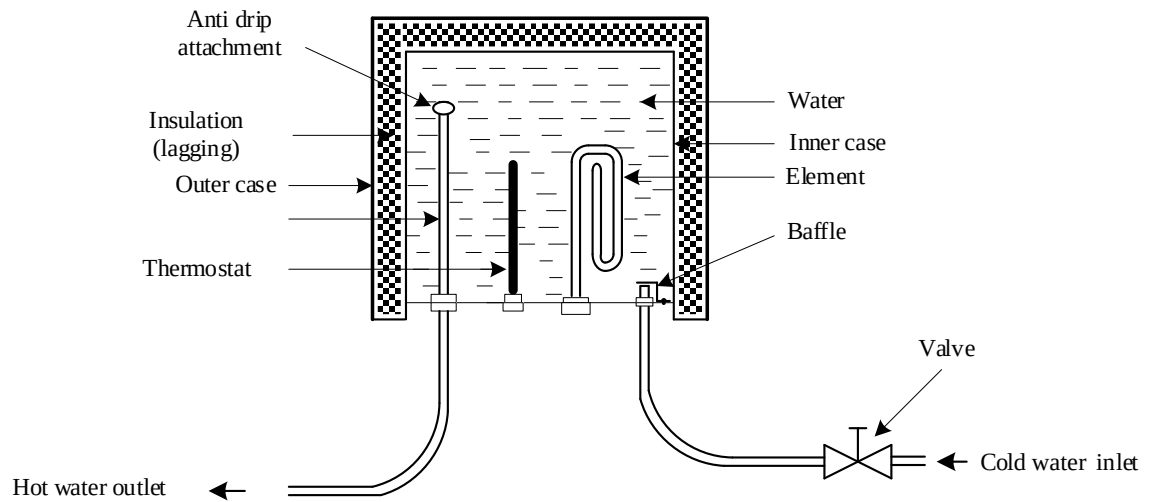


Figure 31 Free outlet water heater

Figure 54 shows the construction details of free outlet water heater. This type of water heater consists of a tinned-copper tank insulated against heat loss by a granulated cork or grass fiber lagging. The inlet valve is fitted with a baffle plate to keep the incoming cold water at the tank. The outlet pipe has an anti-drip syphon which is fitted to prevent the outlet from dripping when the inlet valve is closed. The heating element is composed of a nichrome wire spiral encased in a plated copper tube and insulated with a refractory (heat-resistant) material. The thermostat is used to control the temperature of the water in the tank.

Operation of the Thermostat

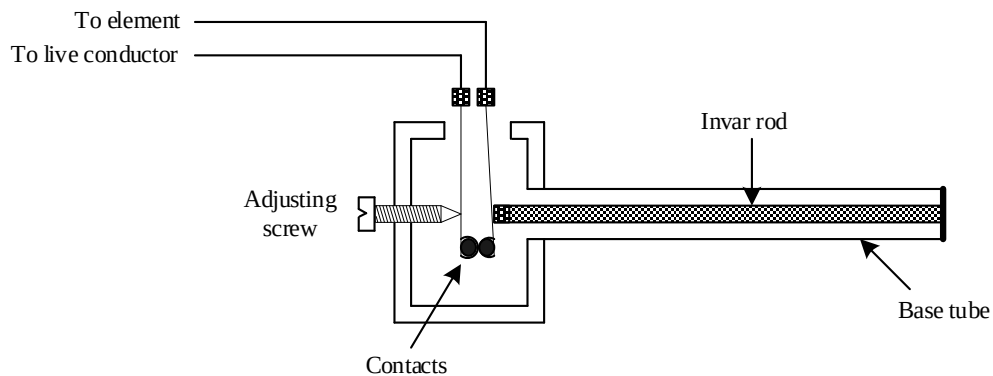


Figure 32 Water heater thermostat

The thermostat must be connected in the phase conductor.

The sequence of operation thermostat is as follows

- The heat from the water heats a brass tube, causing it to expand.

- An invar (non-expanding) rod is fixed at the end of the tube
- The movement of the brass tube draws the invar rod away from the contacts, breaking the heater circuit
- The point at which the contacts open is determined by the pressure on the contacts; an adjustable screw allows variations in the operating temperatures (generally 43°C to 82°C).

Pressure Type Water Heater

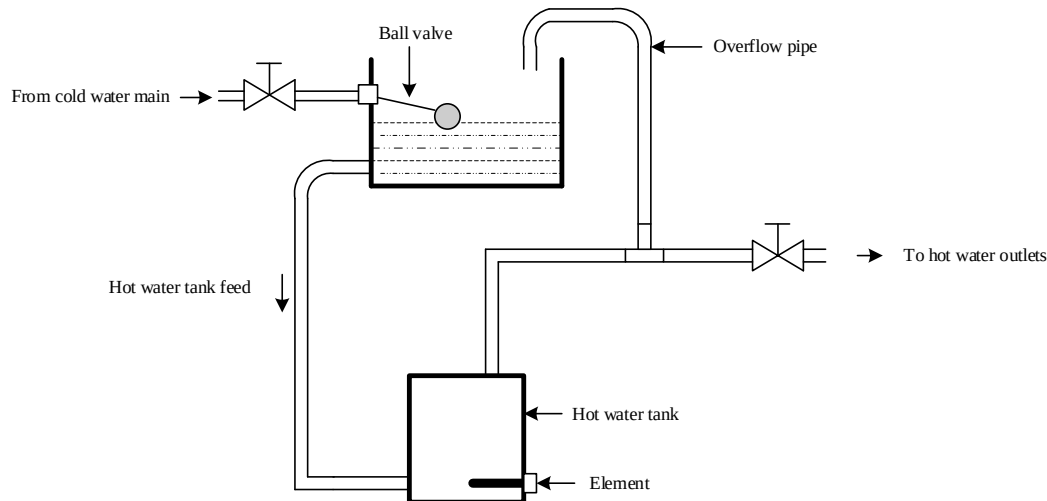


Figure 33 Pressure type water heater

In this type, the hot water tank is fed from a cold water cistern placed above it. The water supplying the cistern is controlled by a ball-valve. When hot water is drawn off the hot water tank (or cylinder) the cistern replenishes it and the ball-valve comes into operation to refill the cistern. In this way a constant pressure (or head) of water is kept on the hot water system.

Difference between pressure and free outlet types.

The free outlet type is controlled by one valve on the inlet side, and the pressure type is designed to supply one or more outlets on the hot water side.

Tank-less water heaters (Instantaneous)

These are water heaters that instantly heat water as it flows through the device, and do not retain any water internally except what is in the heat exchanger coil.

Copper heat exchangers are preferred in these units due to their high thermal conductivity and ease of fabrication.

Advantages

1. Plenty continuous flow of hot water
2. Plenty energy savings

Disadvantage

1. High initial cost

Operation

The heater is normally in the OFF position, but is equipped with flow sensors which will be activated when water flows through them. A feedback loop is used to bring water to the desired temperature. The water circulates through the copper heat exchanger and is heated by gas or electricity. Since there is no storage tank of hot water, the heater provides continuous supply of hot water.

Electric shower heads

In this type, an electric heating element is placed in shower heads to instantly heat water as it flows through. These self-heating shower heads are specialized point-of-use Tank-less water heaters and are widely used in some countries including Kenya.

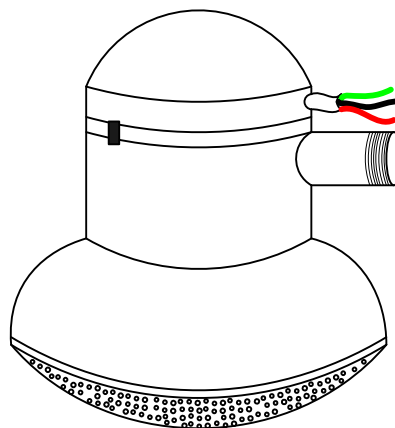


Figure. 34 Electric shower head

Electrical Machines

Electrical machines play an important role in industry as well as in our day to day life. They are used in power plants to generate electrical power and in industry to provide mechanical work, such as in steel mills, textile mills and paper mills

Alternating Current Motors

Motors that are designed to operate on alternating current are called alternating current motors. They convert electrical energy into mechanical energy. Alternating current motors are the most widely used type of motors because, all the generated electrical energy is in the form of alternating current. In addition, ac motors are simple in design, rugged in construction and require little maintenance.

All ac motors basically consist of a stationary part called the stator, a rotating part called the rotor and two end shields that house the bearings the rotor. The design of the rotor and the manner in which current is made to flow through it determines the classification of the motor and its performance characteristics. The stator contains windings which when connected to an alternating current source, creates a rotating magnetic field. At the same time a magnetic field is created in

the rotor by electromagnetic induction which is similar to transformer action. The attraction between the stator and rotor fields causes the rotor field to follow the rotating stator field. AC motors are particularly suitable for constant speed applications. However, variable speed motors are also manufactured. Alternating current motors operate on either single-phase or three-phase supply.

Basic parts of ac motors

All ac motors have similar parts, with the exception of modifications or additions of special parts to a particular type of motor. AC motors essentially consist of:

- (a) Frame: The frame is made of steel and it is the part into which the stator is pressed.
- (b) Stator: The stator is made of a cylindrical core consisting of tightly held together steel laminations, with longitudinal slots around the inner circumference which firmly hold the stator coil windings.

Stator: During the manufacture of some stators, the windings are first wound into the stator slots and properly connected. The wound stator is then dipped into insulating varnish, baked and finally pressed into the frame. Figure 58 shows a completely wound stator and frame of an induction motor before assembly. In other words, the stator is first pressed into the frame and thereafter, individual wires of the coil are wound into the slots, as shown in figure 59. The completely wound stator is then dipped into insulating varnish and baked. Stator windings are made of copper wires

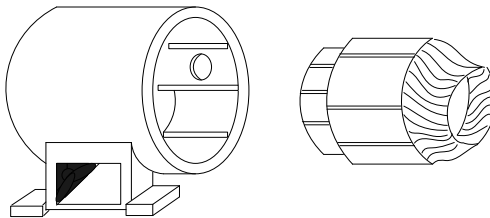


Figure 35. Completely wound stator before the core is pressed into the frame

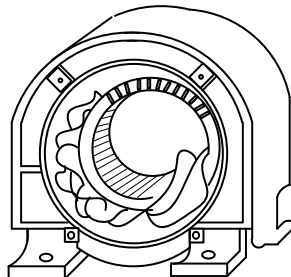


Figure 36. Partially wound stator after the core is pressed into the frame.

Rotor: There are two common types of rotors: the wound rotor and the squirrel-cage rotor. The wound rotor consists of a cylindrical core of steel laminations with longitudinal slots around the circumference. The slots hold coils of insulated copper wires which form the rotor windings. The ends of the windings are connected to slip rings in the three-phase motors and to commutator machines in single-phase repulsion and universal motors.

The squirrel-cage rotor consists of laminated cylindrical steel core with slots around its surface, but instead of holding coils of insulated copper wire, it is embedded with uninsulated large bars of copper or Aluminium conductors. The bars are short –circuited by brazing or welding them to end rings, made of identical material. If the structure formed by the bars and rings were to be viewed outside the core, it resembles a squirrel cage, and hence the name squirrel-cage rotor as illustrated in figure 60. In some rotors, however, the bars and rings are made of one piece of cast Aluminium. The rotor is always mounted on a high-grade steel shaft.

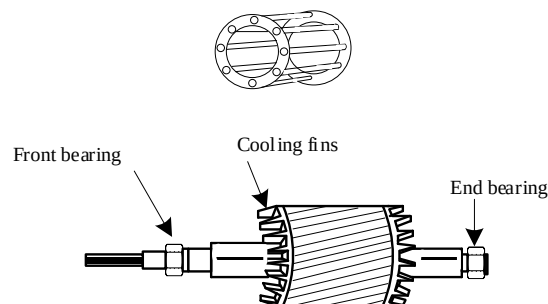


Figure 37 Squirrel-cage rotor

The rotor also carries cooling fans at both ends. These are important because a rise in temperature inside the motor lowers the motor output and in case of excessive temperatures, the insulation of the conductors would burn causing short circuits and fires.

End shields: The two end shields are made of cast steel and bolted to the frame. The end shields house the bearings. The purpose of the bearings is to support the weight of the rotor and to keep the rotor centered within the stator so that as it rotates, friction is minimized and also ensure that it does not strike or rub the stator core.

Single-phase motors

Single-phase ac motors are designed to operate from a single-phase ac supply. They are usually fractional kilowatt motors and are termed small motors. However, some single-phase motors are of 15kw and above and are manufactured for special applications. Single-phase motors are commonly found in house hold appliances such as refrigerators, food mixers, fans, vacuum cleaners and washing machines. They are also found in agricultural, commercial and industrial applications. Single-phase motors are small in size, robust in construction cheap easy to maintain

and particularly useful where only the single-phase ac supply is available. Single-phase motors are divided into three classes:

- i. Synchronous motors
- ii. Induction motors
- iii. Universal motors

The synchronous motor has a constant speed and is not self-starting. There are four types of single-phase induction motors, all named according to the method of starting them. The four in order of their increasing starting torque are: shaded pole, resistance start, capacitor start and repulsion motor. The universal motor also known as series motor, operates on either direct current or single-phase alternating current.

Single-phase synchronous motors.

The single-phase synchronous motor is usually used in timing devices such as wall clocks. It is very small in size and produces a small torque. It has a two-pole stator and the rotor is simply a slotted steel disk or a pile of disks bolted together but insulated from each other.

Operation

The stator contains two poles with their windings connected in series but in such a manner that the poles are of different polarities as shown in figure 55 (a). When single phase current is applied to the stator during the first alteration, the current increases from zero to maximum then falls to zero as shown in figure 61 (b). During the same period the pole pieces are magnetized and say, A becomes a north pole and B a south pole. The magnetic field so produced, varies in strength in similar pattern as the current. During the next alteration, the current changes direction and the polarity of the poles reverses; A becomes a South Pole and B a north pole. The magnetic field is said to be oscillating.

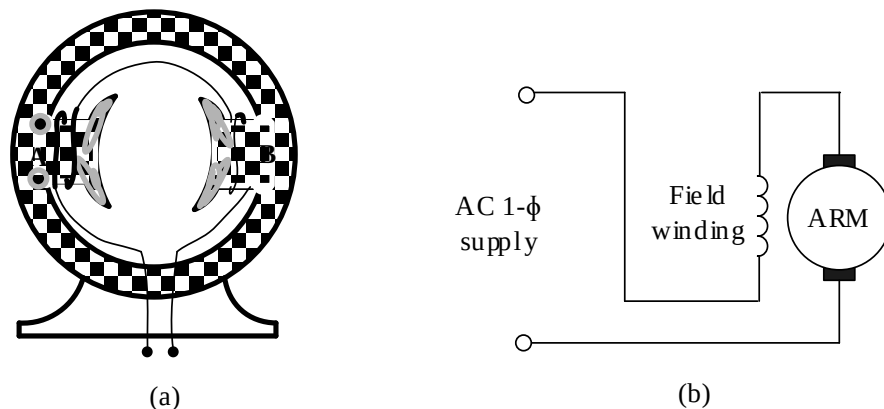


Figure 38 (a) Stator showing the two poles (b) Single- phase series motor

The change in magnetic field induces a voltage and therefore current in the rotor. This process is similar to induction in secondary winding of a transformer. The induced currents in the transformer

create a magnetic field which always opposes the field causing it (Lenz's law). This means that a N-stator pole will produce a N-pole field and a S-pole stator pole, a S-pole field. The polarity of the rotor field changes at the frequency of the alternating current in the stator field. Since these changes are simultaneous, the rotor field pulsates and does not produce a torque. The rotor does not therefore rotate.

If the rotor is caused to rotate by applying an external force, it cuts the magnetic field of the stator. This induces current in the rotor which creates a rotor magnetic field. The pulsating stator magnetic field interacts with the rotor field. Since the induced field is always of the same polarity as the one creating it, and like poles repel, the rotor poles are repelled causing the rotor to rotate. The induced voltage causes induced currents which in turn create a magnetic field in the rotor. This voltage is produced in the following two ways;

- (a) The alternating magnetic field of the stator cuts the rotor conductors and induces the rotor voltage. Since the action is similar to induction of secondary voltage in transformers, this kind of induced voltage is called transformer emf.
- (b) As the rotor rotates, its conductors cut, or are cut by the rotating magnetic field and this induces voltage in the rotor. Voltage induced this way is called speed emf.

It is the combined effect of these alternating emf's which produces the torque that causes the rotor to turn. This applies to all induction motors. The rotor will rotate in either clockwise or anticlockwise direction depending on the direction of the external force applied. Once the rotor has started to rotate, it accelerates until it reaches the synchronous speed. The speed of the rotating field is called the synchronous speed and it depends on the frequency of the supply voltage. The synchronous speed in r.p.m is calculated by multiplying the frequency of the ac input by 60 (number of seconds per minute) and dividing the product by the number of pairs of poles in the motor

$$Speed = 60 \times \frac{\text{frequency}}{\text{number of pairs of poles}} \text{ rpm}$$

Suppose a motor has 8 poles and is connected to 240v, 50Hz line. Its synchronous speed will be;

$$Speed = (60 \times 50) / 4 = 750 \text{ rpm.}$$

The actual speed of the rotor is less than the synchronous speed. If the rotor was to rotate at the same speed as the rotating speed, it would not cut the field, no current would be induced and hence no magnetic field would be created in it, and consequently no torque would be produced. For the rotor to sustain motion, there must be some difference between the synchronous speed and the actual speed. The difference in speed is called SLIP. Slip is expressed as a percentage of synchronous speed. In single-phase synchronous motors, it is less than 4% and for practical purposes, the motor is considered to rotate at synchronous speed.

Resistance- start induction run motor (split-phase)

The resistance start, induction run motor is also referred to as split-phase motor. The main parts of the motor are frame, stator, squirrel-cage rotor, two end shields and a centrifugal switch mounted on the rotor shaft. Figure 62 shows parts of a disassembled resistance-start induction run motor.

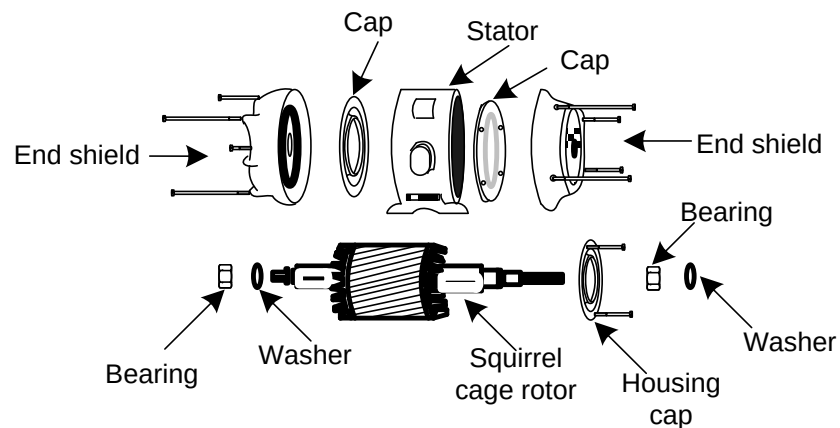
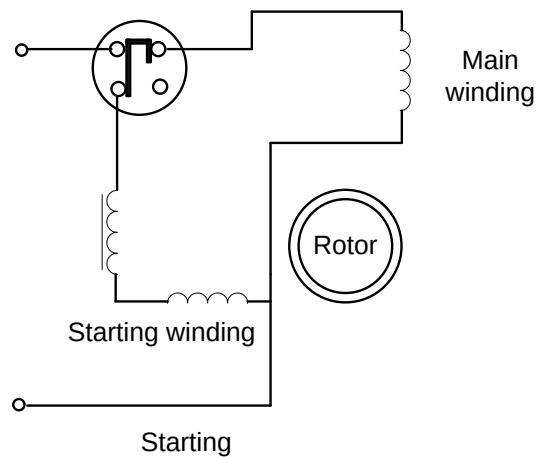


Figure 39 Parts of a disassembled resistance-start induction-run motor.



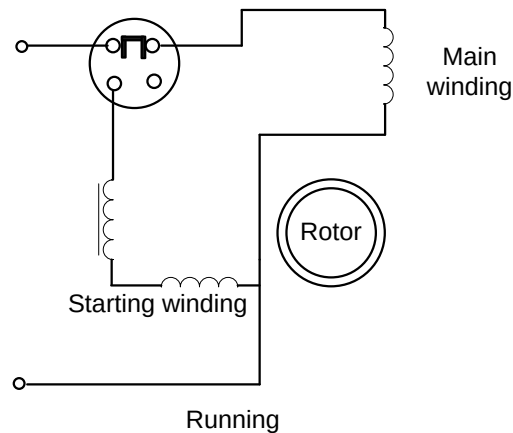


Figure 40. Split-phase motor.

Capacitor Start Motors.

There are two types of capacitor start motors, namely the capacitor start-induction run and capacitor start-capacitor run. The main difference between the two is that in the former, the starting winding and the capacitor are disconnected from the circuit when the motor attains 75% of the rated speed, while in the latter, the starting winding is permanently connected in the circuit.

Capacitor start-induction run motor

In this type of motor, a capacitor is connected in series with the starting winding as shown in figure 64. An electrolytic capacitor is commonly used.

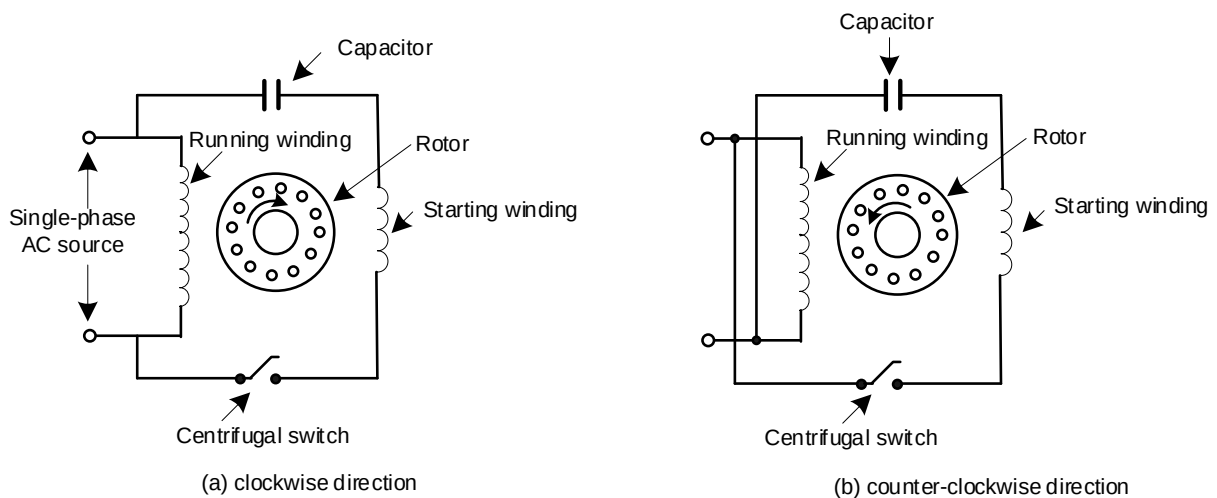


Figure 41. Connection of a capacitor-start induction-start run motor direction.

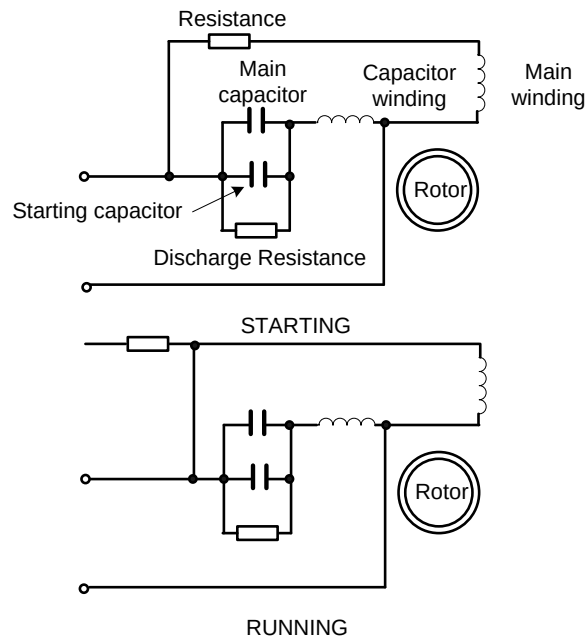


Figure 42. Single-phase capacitor start induction run.

Electrolytic capacitors have short duty service, and are easily damaged by frequent surge currents. Therefore, the capacitor start-induction run motor is suitable for situations which require relatively few starts in a short period of time. Many manufacturers recommend that the motors be started less than 20 times in an hour.

Operation

The running winding of this motor is embedded in iron and therefore highly inductive. A capacitor of suitable size connected in the starting winding, causes a considerable phase difference between the currents in the two windings. The phase difference between these two currents gives the motor a high starting torque. It runs efficiently with fairly good power factor and its starting torque can be as high as three times its full load torque. It is essentially a constant speed machine and is used for light industrial work, office and domestic applications where better starting torque is required.

Capacitor start - Capacitor run motor

(i) Single-Capacitor Motor

This motor is known as capacitor start-capacitor run because the starting winding is left in the circuit during running. Figure 66 shows this type of motor with the centrifugal switch omitted meaning the motor continues running as a two phase machine. The running of this motor is smoother with good power factor but the capacitor and running winding be continuously rated. This motor is used where a high starting torque is not essential so that a lower value capacitor

would be used for running continuously. It is mostly used in ventilating fans, small pumps, circulating pumps in central heating systems etc.

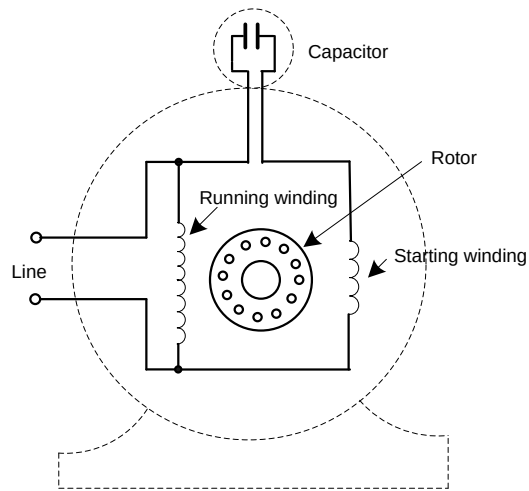


Figure 43. Connections in a single value capacitor-start, capacitor run motors

(ii) Two Value Capacitor Motor

This motor has two capacitors connected in parallel, and both connected in series with the starting winding as illustrated in figure 67. Usually an oil capacitor is connected permanently while an electrolytic capacitor is connected through a centrifugal switch. The motor starts with both capacitors in parallel and the switch opens the electrolytic capacitor when the motor has reached 75% of the rated speed. The motor then runs with the oil capacitor alone in series with the starting winding. This motor is a smooth running machine with high frequency, good power, and high starting torque with quiet running. It is used for light industries and office machines.

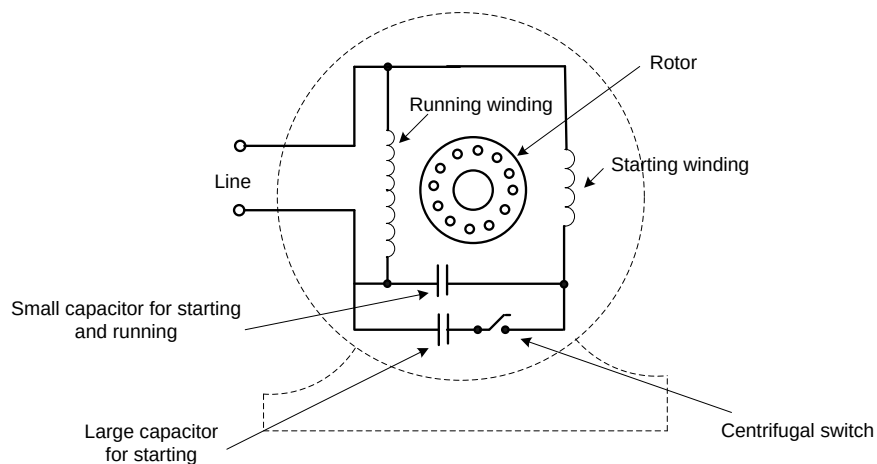


Figure 44. Internal connection of a two-value capacitor motor

Repulsion motors

Repulsion motors derive their names from the fact that they produce the torque required for rotation from the repulsion of two like poles. These motors have relatively high starting torque and draw less starting current than the other types of single phase motors.

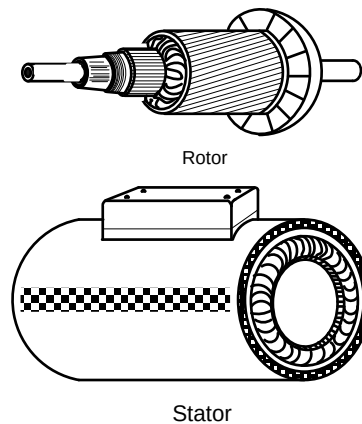


Figure 45. The main parts of a repulsion motor

Figure 68 illustrates the essential parts of a repulsion motor. These parts are;

- (a) The stator, which has only one field winding similar to the running winding of single-phase motors. It usually contains four, six or eight poles.
 - (b) The rotor, which is the wound type. It has insulated copper windings and its terminations are connected to a commutator. Brushes ride on the commutator and short circuit the windings. Note that the rotor is not electrically connected to the ac source.
 - (c) The two end shields that house the bearings which support the rotor and the shaft
 - (d) The frame, whose purpose is to securely hold the stator core to which end-shields are bolted.
- The construction of repulsion motors differs from split phase motors in that they have a wound rotor and the stator has only one type of field winding

Types of repulsion motors

There are three types of repulsion motors; these are the repulsion start- repulsion run motor, repulsion-start induction run and the repulsion induction motor. They are all reversible and have excellent starting torque.

- (a) *Repulsion Start-Repulsion run motor*. This is the basic repulsion motor. Its speed can be varied by changing the value of the supply voltage. If all the mechanical load is removed, the motor

will accelerate to a dangerous high speed called runaway speed. It is therefore used on permanent loads such as elevators.

- (b) *Repulsion Start-Induction run motor.* This motor has a centrifugal device in the rotor. When the motor has attained approximately 75% of its rated speed, the centrifugal device passes a short-circuiting ring into the commutator, which short-circuits the commutator segments and current no longer flows through the brushes. The motor then runs as an induction motor. The motor has excellent starting torque and is reversible. The motor is used in commercial refrigerators, compressors and pumps.
- (c) *Repulsion-Induction motor.* This type of repulsion motor has a squirrel cage winding on the rotor under the regular wound winding, as shown in figure 69. The added winding is inductive and serves to limit the no-load speed when the motor speeds above the synchronous speed. The motor has fairly constant speed from no-load speed because of the squirrel-cage winding. The motor does not have a centrifugal switch and therefore needs little maintenance and repairs. It starts as a repulsion motor and runs as an induction motor.

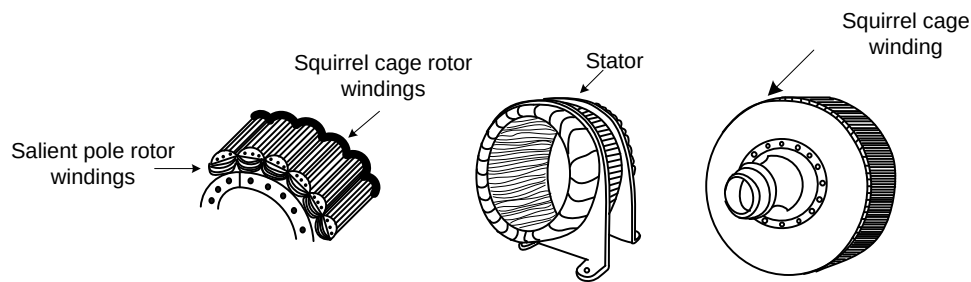


Figure 46 Details of the rotor of a repulsion-induction motor

Shaded pole motor

The shaded pole motor is a fractional kilowatt motor. It belongs to the group of split-phase motors. The motor does not have the regular starting winding, centrifugal switch or starter mechanism. The motor has a squirrel cage rotor. However, a short circuit ring is mounted around about a third of the pole area of each stator pole. The area covered by the ring is called the shaded pole and the ring is called the shading coil. All the poles are shaded on the same side, and the shading coil is made of low resistance copper. Figure 70 shows a typical shaded pole motor.

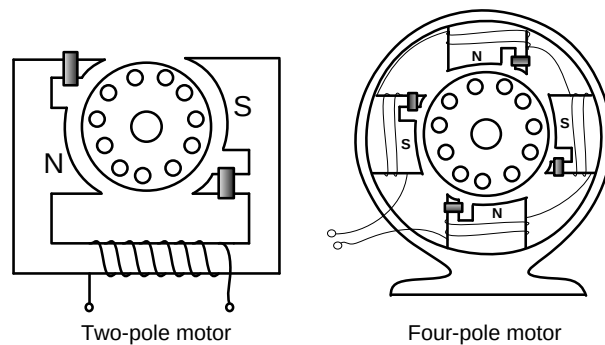


Figure 47 Shaded pole motor

Operation

The shading coil causes the building up and the collapse of magnetic field around the shaded pole to lag by opposing the change of current through it. This creates a shifting magnetic field along the surface of the pole.

UNIVERSAL MOTOR

Universal motor, also known as the series motor, is designed to operate on either single-phase alternating current or direct current. Figure 71 illustrates the main parts of the motor.

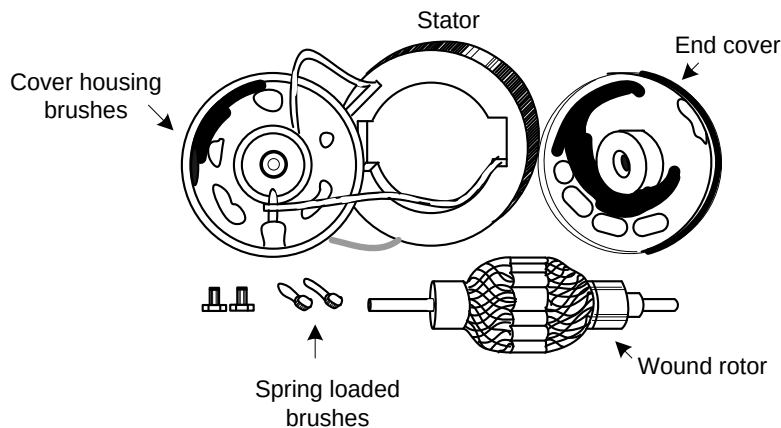


Figure 48 the main parts of a universal motor

The frame is made of cast Aluminium, cast iron or rolled steel. Two end shields, made of steel, serve the same purpose as in other motors. The stator is made of steel laminations securely held together and bolted to the frame. The rotor consists of a cylindrical core of steel laminations with longitudinal slots into which insulated copper windings are wound. The windings are wound in such a way that they create two opposite poles in the stator. The rotor windings are connected to

commutator segments. Two carbon brushes ride on the commutator, and are connected to the field winding.

Operation

Stator and rotor windings are connected in series as shown in figure 72.

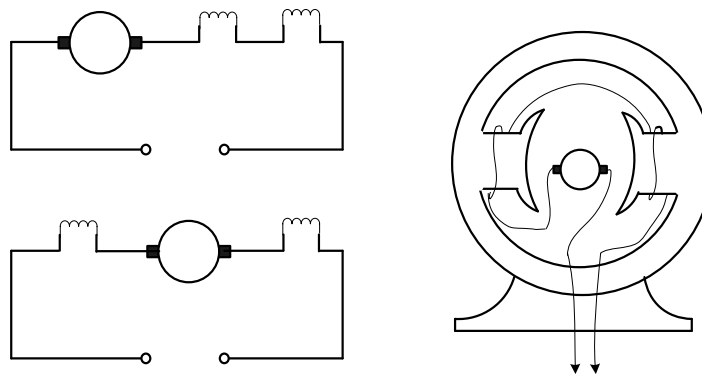


Figure 49 Series connected of field and rotor windings of a universal motor

When the motor is connected to single phase ac supply, the same amount of current flows through both the field and rotor windings because they are in series. By the method of connections, the magnetic field set up in the rotor will have the same polarity as the poles close to it. The two like poles will repel and produce torque, and the rotor therefore rotates, this is similar to torque production in a repulsion motor.

Universal motors are variable speed motors. The speed depends on the voltage applied and the load. The motor has very high starting torque. They are widely used in portable commercial such as saws, drills, grinders and sanders, domestic appliances such as food mixers, sewing machines and vacuum cleaners. They usually cause radio and TV interference because of sparking at the brushes. To obtain variable speeds, control rheostats, impedance coils tapped windings or centrifugal devices are used. The motors may achieve excessive high speeds and therefore they are normally started and operated with the load connected. The direction of rotation is changed by reversing either the field or armature connections